



Linear Control Systems 25411

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Assignment 6

Fall 1404

Due Date: 1404/11/1

Note 1: *purple* problems are bonus ones.

1 Nyquist Diagram

For each transfer function sketch the full Nyquist diagram. Then using the Nyquist criteria discuss upon the number of unstable closed-loop poles under the unity feedback assumption.

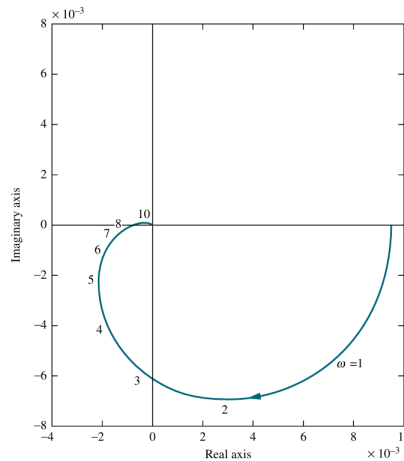
a) $G_1(s) = \frac{1}{s(s+2)(s+4)}$

b) $G_2(s) = \frac{s+5}{(s+2)(s+4)}$

c) $G_3(s) = \frac{s+3}{s^4}$

2 Transfer Frequency Information

Draw the separate approximate magnitude and phase curves from the polar plot given below.



3 Nyquist Stability Criterion

Using the Nyquist criterion, find the range of K to insure the closed-loop stability under unity feedback assumption. Verify your answer using the Routh-Hurwitz criterion.

$$G(s) = \frac{K}{s(s+2)(s-1)}$$

4 Riderless Bicycle Stability Analysis

A riderless bicycle can be stabilized using feedback control from the measured lean angle to the steering mechanism. The open-loop dynamics (from steering input to lean angle output) at constant forward velocity is modeled as:

$$G_p(s) = G_1(s)G_2(s) = \left[3.373 \frac{s+8}{s^2-18.9} \right] \left[e^{-0.015s} \frac{(33.9)^2}{s^2+40.68s+(33.9)^2} \right]$$

A PID controller has been designed as:

$$G_c(s) = 2.5177 \left(1 + \frac{1.5431}{s} + 0.075s \right)$$

Assume unity negative feedback.

- Using MATLAB, plot the Nyquist diagram of the open-loop transfer function $L(s) = G_c(s)G_p(s)$. (Hint: Use `pade` to approximate the time delay or handle it directly with `nyquist`.) Verify the closed-loop stability by applying the Nyquist stability criterion. Determine the number of encirclements of the point -1 and the number of open-loop unstable poles.
- Using MATLAB, plot the magnitude $|L(j\omega)| = |G_c(j\omega)G_p(j\omega)|$ versus frequency (Bode magnitude plot of the loop gain). Determine the gain crossover frequency ω_c (where $|L(j\omega_c)| = 1$).
- From the magnitude plot, find the minimum value of $|L(j\omega)|$ for $\omega \leq 0.1\omega_c$ and the maximum value of $|L(j\omega)|$ for $\omega \geq 10\omega_c$.

Hint: Use Control System Toolbox functions such as `tf`, `pade`, `series`, `nyquist`, `bode`, and `margin`. The plant has open-loop unstable poles.

5 Time Delay Approximations

Approximating the time delay e^{-sT} using a rational function is essential for applying standard control techniques. One method splits the exponential (not Padé):

$$e^{-sT} = \frac{e^{-sT/2}}{e^{sT/2}}$$

and approximates each part with Taylor series.

The Padé approximation is defined as:

$$e^{-sT} \approx R_{m,n}(Ts) = \frac{\sum_{i=0}^m p_i(Ts)^i}{\sum_{i=0}^n q_i(Ts)^i}, \quad p_0 = q_0 = 1$$

with coefficients

$$p_i = (-1)^i \frac{(m+n-i)!m!}{(m+n)!i!(m-i)!}, \quad q_i = \frac{(m+n-i)!n!}{(m+n)!i!(n-i)!}.$$

- Assume $T = 1.5$. Using MATLAB, compare the Bode diagrams of the Taylor series approximations of orders $n = 1, 2, 3, 4, 5$. Plot all on the same figure.
- Compare the step responses of these Taylor approximations. Comment on stability and initial behavior.
- Derive the coefficients p_i, q_i for the Padé approximation $R_{2,2}(Ts)$. Are they identical to the Taylor series coefficients up to order 4?
- Using MATLAB with $T = 1.5$, compare Bode diagrams of the balanced Padé approximations $R_{1,1}$, $R_{2,2}$, $R_{3,3}$, and $R_{4,4}$.

- e) Compare their step responses. Explain the discontinuity at $t = 0$ for equal-order Padé.
- f) Compare Bode diagrams and step responses of the unbalanced approximations $R_{1,2}$, $R_{2,3}$, and $R_{3,4}$.
- g) Explain why equal-order ($m = n$) Padé is preferred for frequency-domain analysis while lower numerator order ($m < n$) is used for time-domain simulations.

Hint: Use `tf`, `pade`, `bode`, `step`, and symbolic computation if needed for coefficients.

6 Controller Design using Phase-Lag Compensator

In class, the design of phase-lag compensators using frequency domain methods was discussed. Consider the speed control system of a supervisory vehicle as follows, where $T_d(s)$ represents the disturbance due to terrain irregularities. In this system, a phase-lag controller structure with specified pole and zero locations is used. Tune the gain K to achieve a trade-off between acceptable percent overshoot and steady-state error using the Nichols chart.

$$G(s) = \frac{10}{(s+1)(s^2+2s+4)}, \quad G_c(s) = K \frac{s+2}{s+1}$$

Assume unity negative feedback.

- a) Derive the steady-state error e_{ss} for a unit step input in terms of K .
- b) Assuming $K = 20$, compute e_{ss} and verify it is less than 9% of the input amplitude.
- c) For $K = 20$, compute the loop gain $L(s) = G_c(s)G(s)$ and use the Nichols chart to find the peak response M_p in dB and phase margin $P.M.$. Then, estimate the damping ratio ζ and percent overshoot.
- d) Assume a maximum overshoot of 25% is acceptable, corresponding to $\zeta \geq 0.4$ and $M_p \approx 2.67$ dB. Adjust K by shifting the Nichols chart downward and find the new K . Compute the corresponding e_{ss} .
- e) Write a MATLAB code that, for a given K , computes and plots the step response using `tf` and `step` commands. Also, extract and display percent overshoot, settling time, peak time, and e_{ss} .
- f) Use a `for` loop to compute and tabulate the performance metrics (overshoot, settling time, peak time, e_{ss}) for $K = [4.44, 10, 20]$. Compare with the expected values from the Nichols chart.
- g) For $K = 10$, compute the phase margin and bandwidth (ω_B at -3 dB) using Bode plot. Validate the settling time using the approximation $T_s \approx 4/(\zeta\omega_n)$ and the relation $\omega_B \approx -1.19\zeta + 1.85\omega_n$.
- h) **Simulate the step response for the values of K in part f and plot them together. Comment on the trade-off between transient response and steady-state error.**

Hint: Use Control System Toolbox functions like `tf`, `feedback`, `stepinfo`, `bode`, `margin`, and `nichols`. For Nichols chart analysis, you may use `nichols` to plot and manually interpret or approximate.

7 Nyquist Stability Criterion for Time-Delayed Systems

In class, the Nyquist plot of time delayed systems was discussed. Assume that we wish to analyze the stability of the following simple time-delayed system using the Nyquist criterion, despite the fact that time delays introduce infinite-dimensional dynamics.

$$L(s) = K \frac{e^{-Ts}}{s+P}$$

- a) Derive the conditions (magnitude and phase) under which the Nyquist plot passes through the critical point -1 for marginal stability.

- b) Unfortunately, equations in part a are transcendental. Write a MATLAB code which for a known proportional gain K , plots the Nyquist diagram (assuming unity negative feedback). $T = 0.2$ and $P = 1$.
- c) Use a `for` loop to plot Nyquist diagrams as K varies in range of $0 : 0.1 : 20$. Observe how the plot scales with K .
- d) Sketch the Nyquist plot for a value of K where it touches -1 . Note that for delayed systems, the plot spirals inward due to the phase lag from the delay, potentially leading to multiple crossings.
- e) Respecting the obtained Nyquist plots, compute the value of K which causes the onset of instability (when the plot starts encircling -1).
- f) For the value of K obtained in part e, what is your expectation from the Bode plot? Sketch it using `bode` command and relate the phase margin to stability.
- g) Simulate step response of the system regarding the obtained value of K in part e and show that the system is on the verge of instability.

Hint: To do part b, use `tf` with 'InputDelay', T or `pade` for approximation if needed. The `nyquist` command supports delays. For encirclements, you may need to inspect visually or use advanced functions.

8 Root-locus of Time Delayed Systems

In class, the Nyquist plot of time delayed systems was discussed. Assume that we wish to draw the root-locus of following simple time-delayed system, despite the fact that the MATLAB `rl locus` command does not support time-delayed systems.

$$L(s) = \frac{e^{-Ts}}{s + P}$$

- a) Derive corresponding conditions that a desirable point in s-domain belongs to the root-locus plot of $L(s)$.
- b) Unfortunately, equations in part a are nonlinear. Write a MATLAB code which for a known proportional gain K , calculates the associated closed-loop poles (assuming unity negative feedback). $T = 0.2$ and $P = 1$.
- c) Use a `for` loop to compute closed-loop poles as K varies in range of $0 : 0.1 : 20$.
- d) Sketch the root-locus. Note that for some values of K we might have more than one root! Actually since the equations contains sine and cosine functions, in some cases we might have an infinite number of roots. This property of time-delayed systems is interesting.
- e) Respecting the obtained root-locus, compute the value of K which causes instability.
- f) For the value of K obtained in part e, what is your expectation from Nyquist plot? sketch it using `nyquist` command and validate your expectation.
- g) Simulate step response of system regarding the obtained value of K in part e and show that system is on the verge of instability.

Hint: To do part b, it might be helpful to use `syms`, `solve` and `subs` commands.